An introduction to the fabulous world of LeRobot

Grenoble Data Science Meetup

2026-10-29

Emmanuel Virot

emmanuel.virot@wormsensing.com



← In 2025, presentation of the robotic arm SO-101
Source: Hugging Face

LeRobot

Hugging Face

worms

Outline

1- Introduction to LeRobot

The ChatGPT moment of robotics

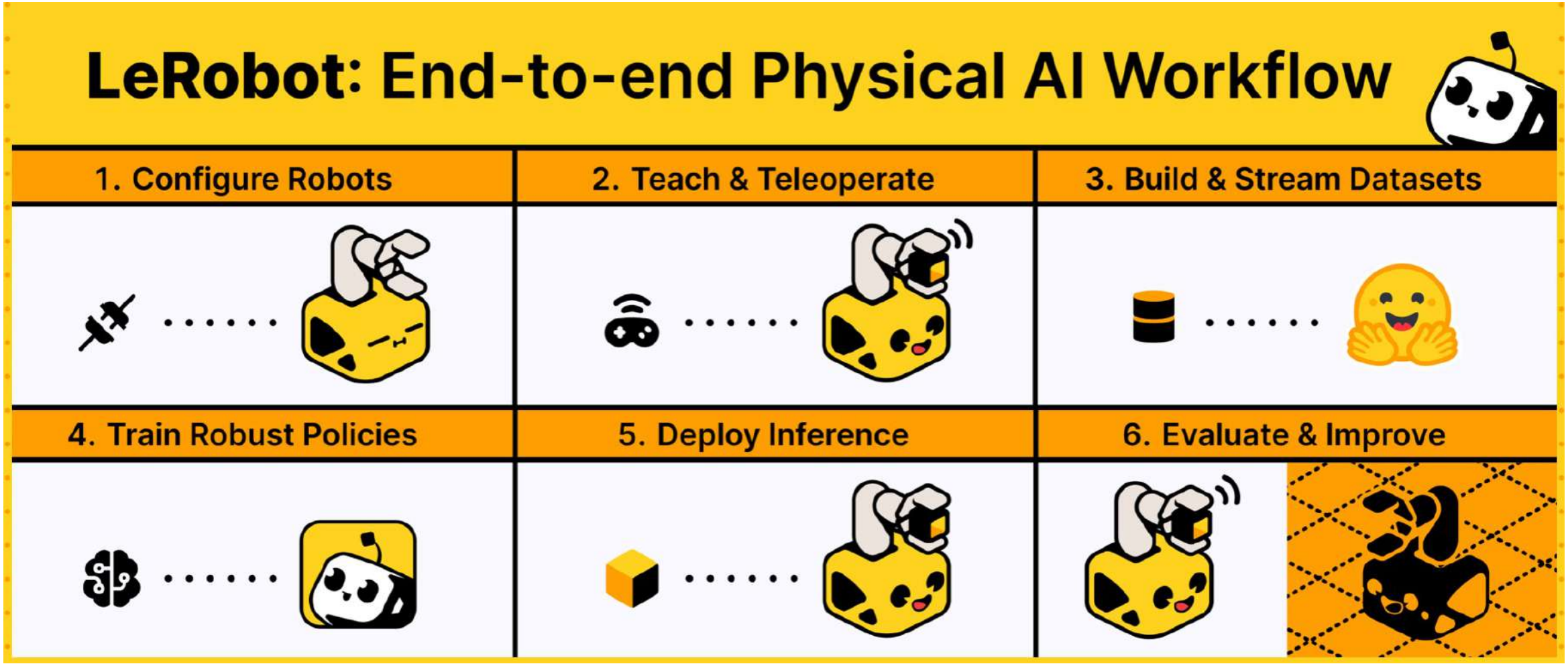
- The robots
- Recording datasets
- The community
- Training models
- Inferences
- The power of hackathons

2- Adding tactile sensing in learning-based robotics

Humanoid robots can do backflips,
but they can’t reliably pick up a screwdriver

- The missing piece in robotics
- Simple tactile integration in vision-based models
- SpectRobot

LeRobot pipeline



Infographic credits: Hugging Face and Nikodem Bartnik

LeRobot pipeline

https://huggingface.co/docs/lerobot/

GET STARTED

How to get started?

TERMINAL

```
# Beginner mode
pip install lerobot

# Advanced mode
git clone git@github.com:huggingface/lerobot.git
cd lerobot
pip install -e ".[many-cool-options]"
```



And let's go!

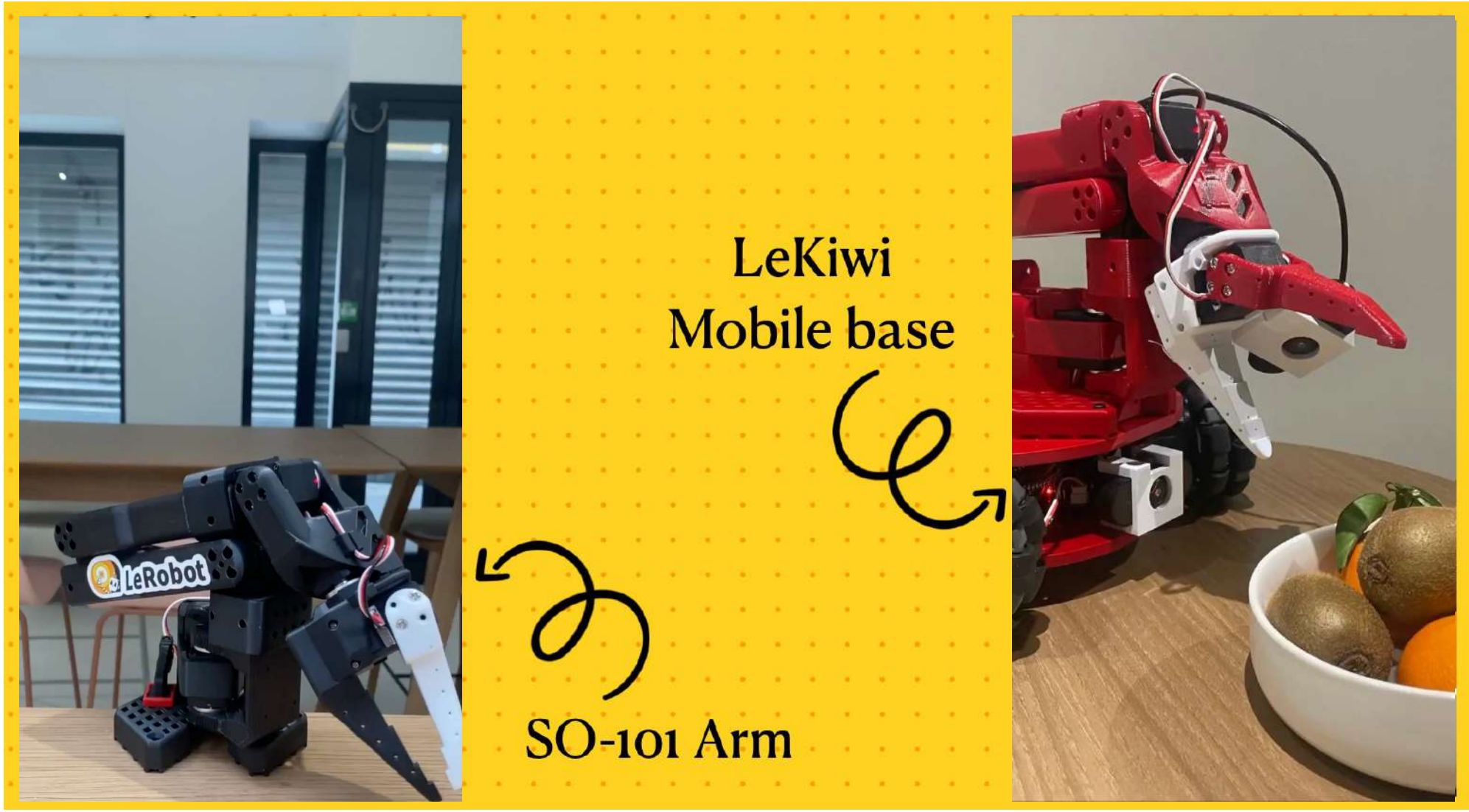
Infographic credits: Hugging Face and Nikodem Bartnik

The robots that are supported



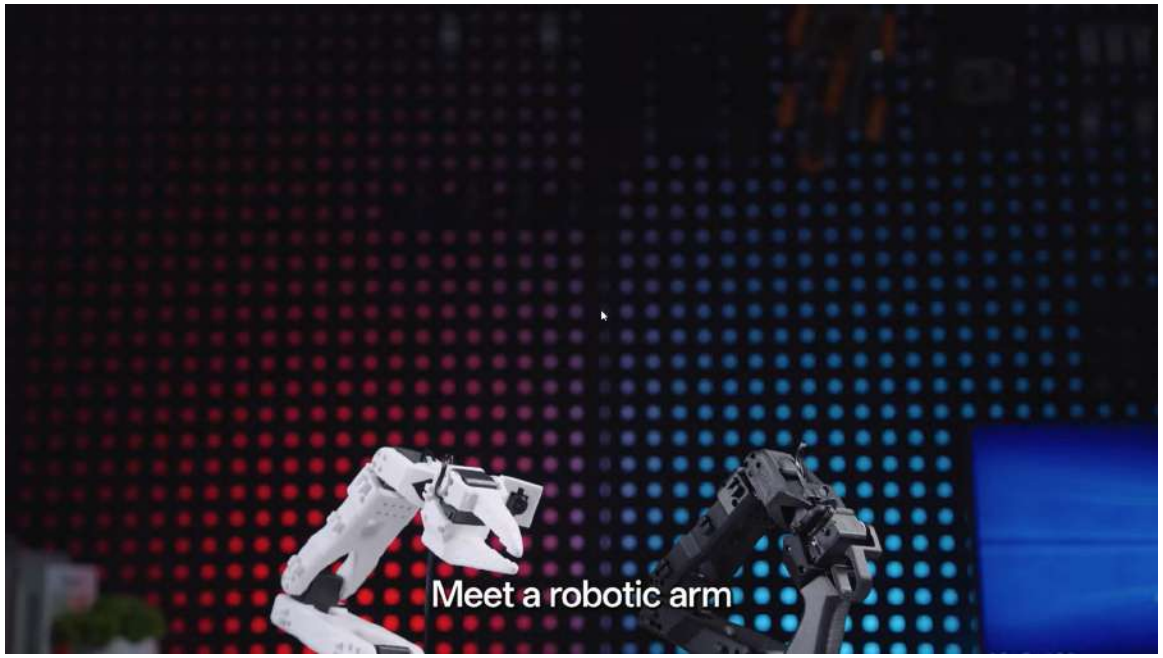
Infographic credits: Hugging Face and Nikodem Bartnik

The robots that are supported



Infographic credits: Hugging Face and Nikodem Bartnik

How to get a SO-101 robotic arm?





It costs around 300 € for a leader arm + follower arm (instead of 10,000 € in the past...)
+ a huge community for free, ready to help

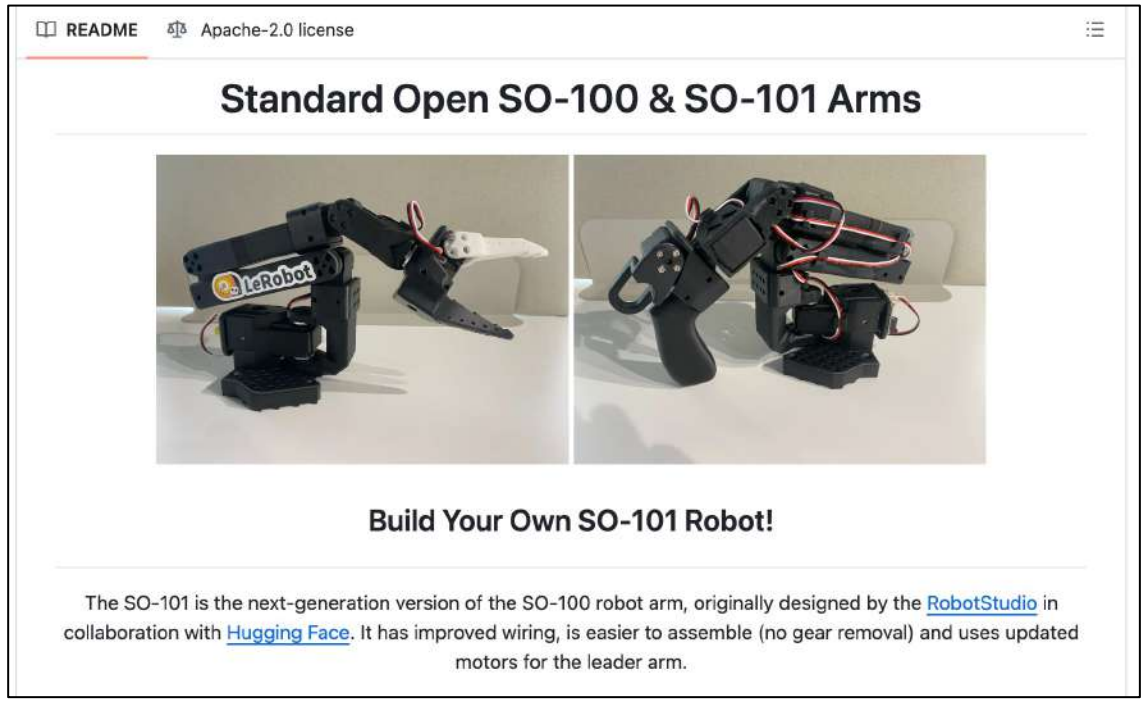
Video: https://www.youtube.com/watch?v=oitT8geMat0

How to get a SO-101 robotic arm?

The vanilla way: built it from scratch!

https://github.com/TheRobotStudio/SO-ARM100

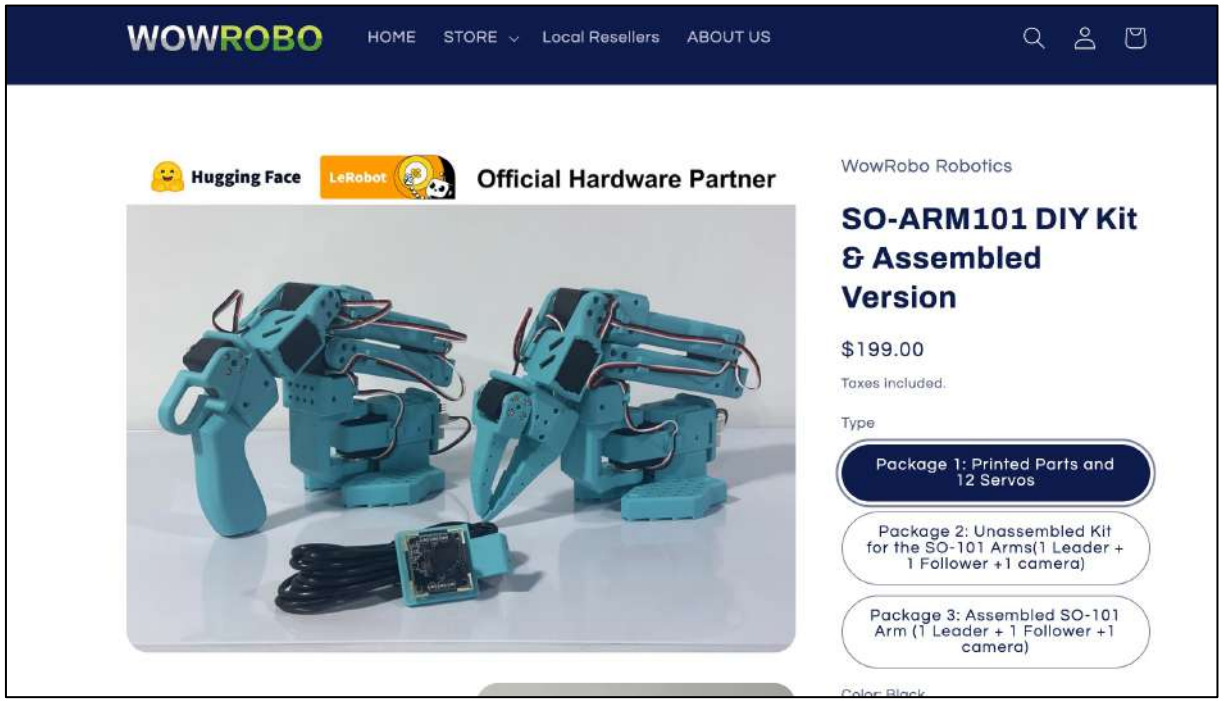
- Shopping list (Amazon, AliExpress, ...) ~100 €
- STL files for your 3D printer



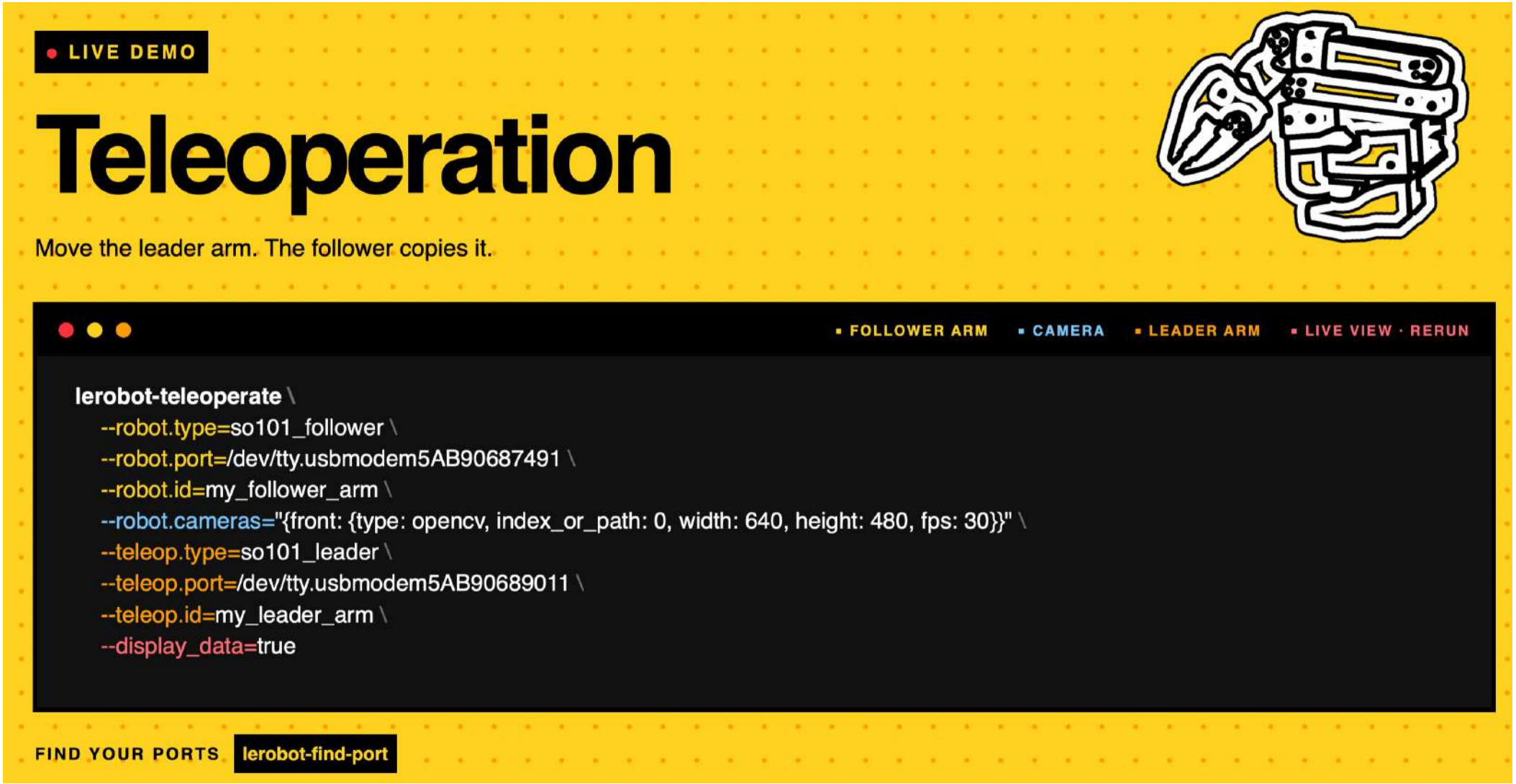
The safe way: external vendors

Example: https://shop.wowrobo.com/

- Assembled and calibrated ~300 €



Recording datasets



Infographic credits: Hugging Face and Nikodem Bartnik

Recording datasets



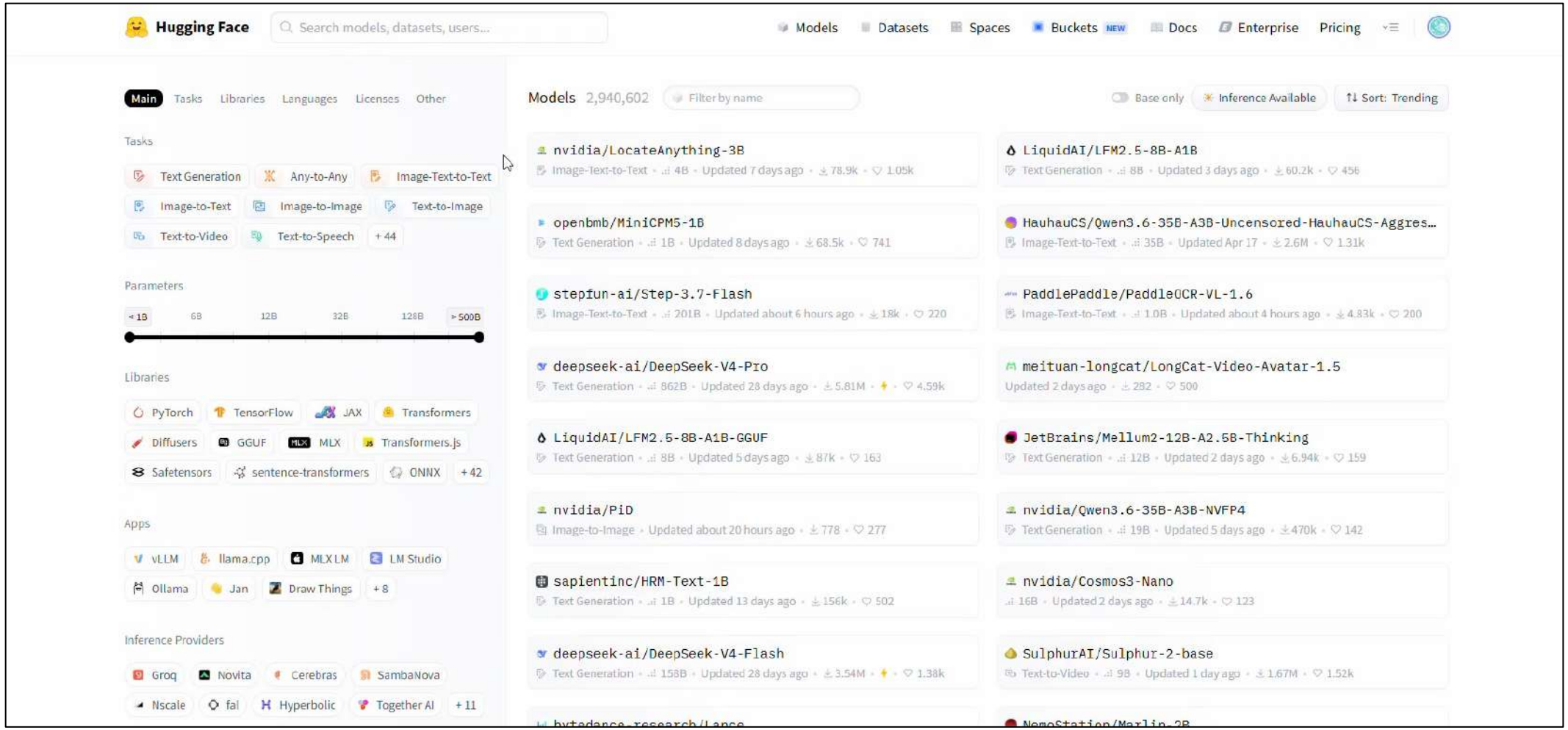
Infographic credits: Hugging Face and Nikodem Bartnik

Recording datasets



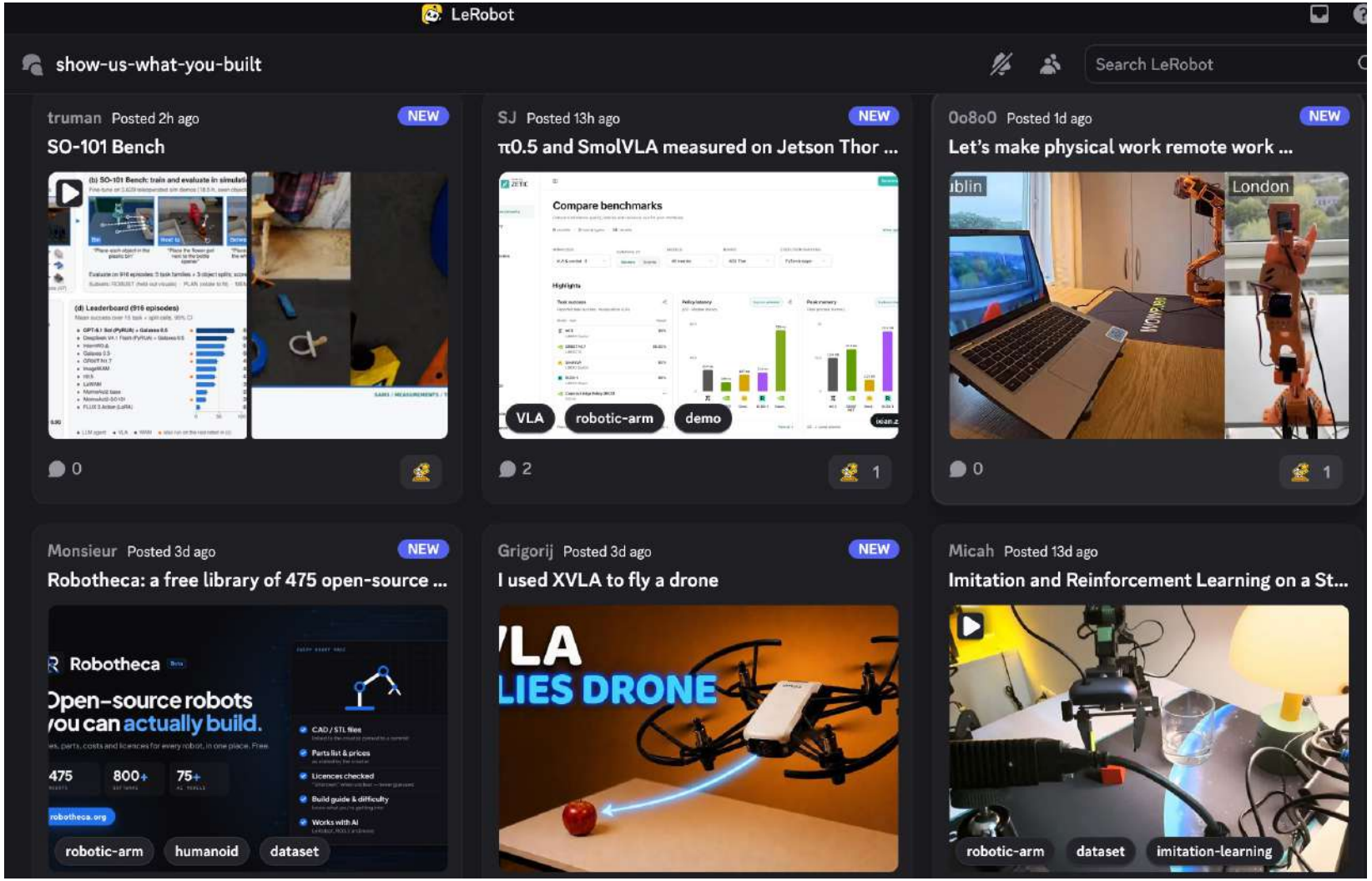
Infographic credits: Hugging Face and Nikodem Bartnik

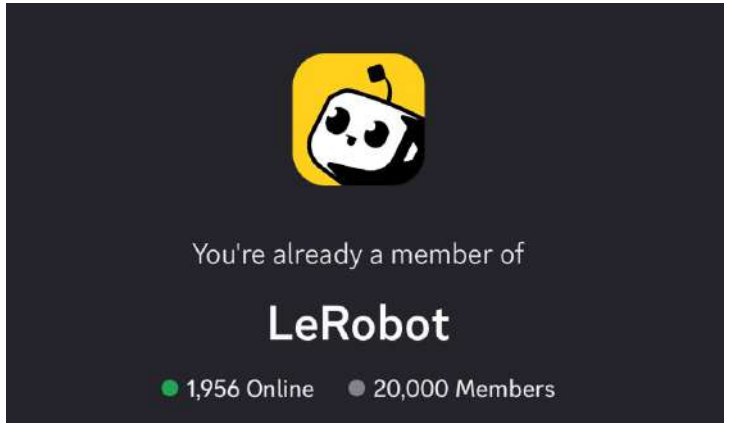
The community: the Hugging Face hub



https://huggingface.co/spaces/lerobot/visualize_dataset?path=%2Fjogarulfop%2F2026-04-30_miscelanious_tasks_usb_chalk_magnet_glass%2Fepisode_0%3Ft%3D7

The community: Discord

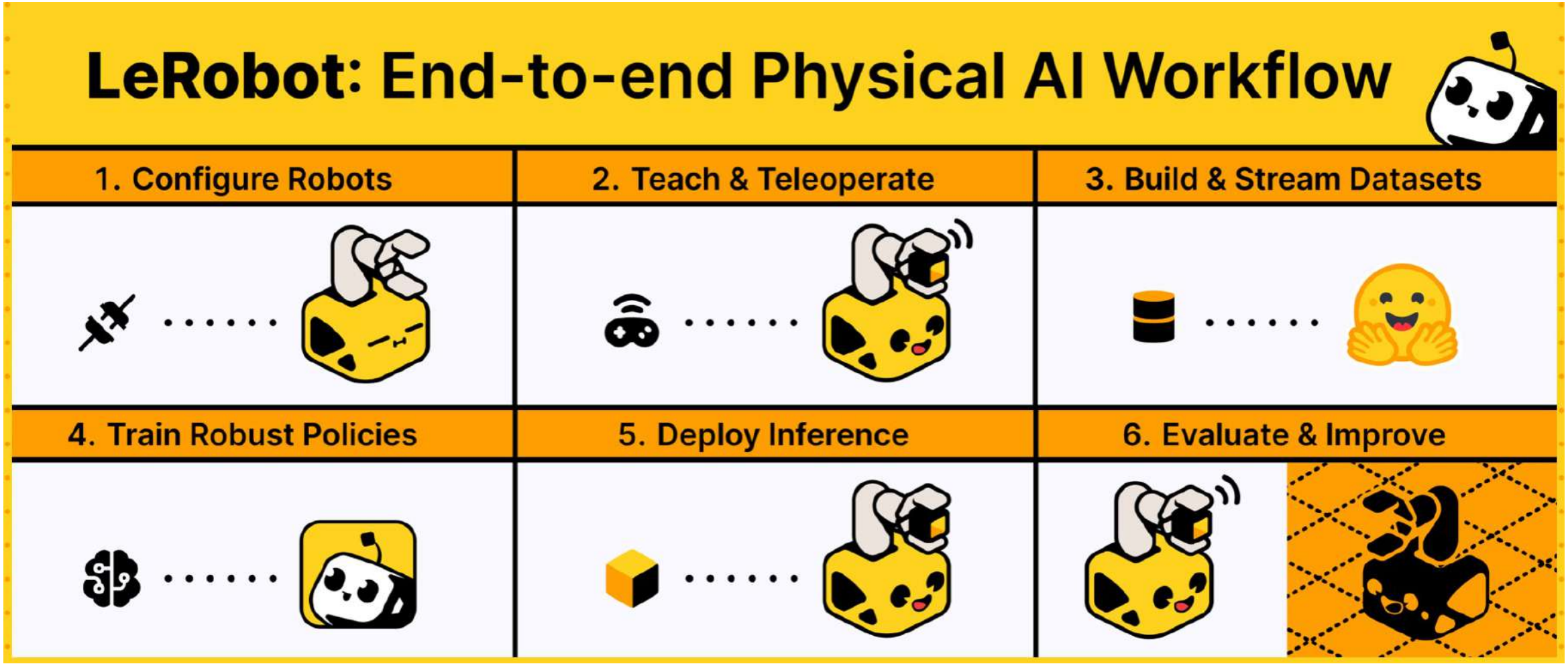




A huge community (20,000+ members) ready to help out!

discord.gg/q8Dzzpym3f

LeRobot pipeline



Infographic credits: Hugging Face and Nikodem Bartnik

Training models (selection)

80 millions parameters (considered “light”)
~6 Go VRAM for training
~2 Go VRAM inferences

not pretrained (expect the ResNet encoder)

Imitation Learning: ACT (Action Chunking with Transformers, from the Stanford ALOHA project), Diffusion (from Columbia and Toyota Research Institute), …

Vision-Language-Action (VLA) models: Pi0, Pi0Fast, Pi0.5 (Physical Intelligence), GR00T N1.7 (from NVIDIA),
SmolVLA (from Hugging Face and specifically trained on SO-101), …

3 billions parameters
~40 Go VRAM for training
~16 Go VRAM for inferences

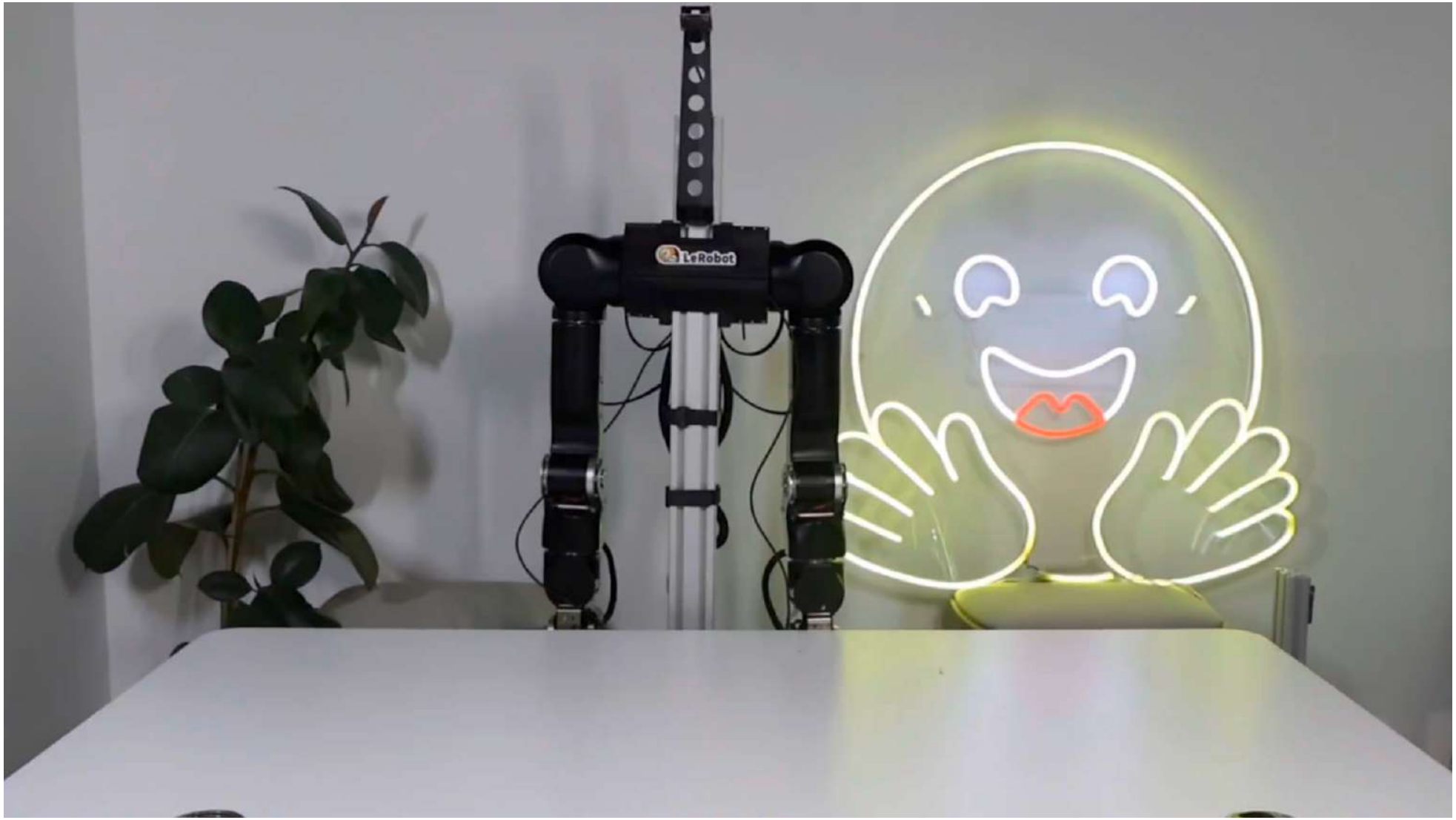
pretrained (fine tuning)

World Models: VLA-JEPA (the JEPA idea itself is well known because of Yann LeCun’s work at Meta), FLUX 3 Action (Black Forest Labs), …

3 billions parameters
~80 Go VRAM for training
~16 Go VRAM for inferences

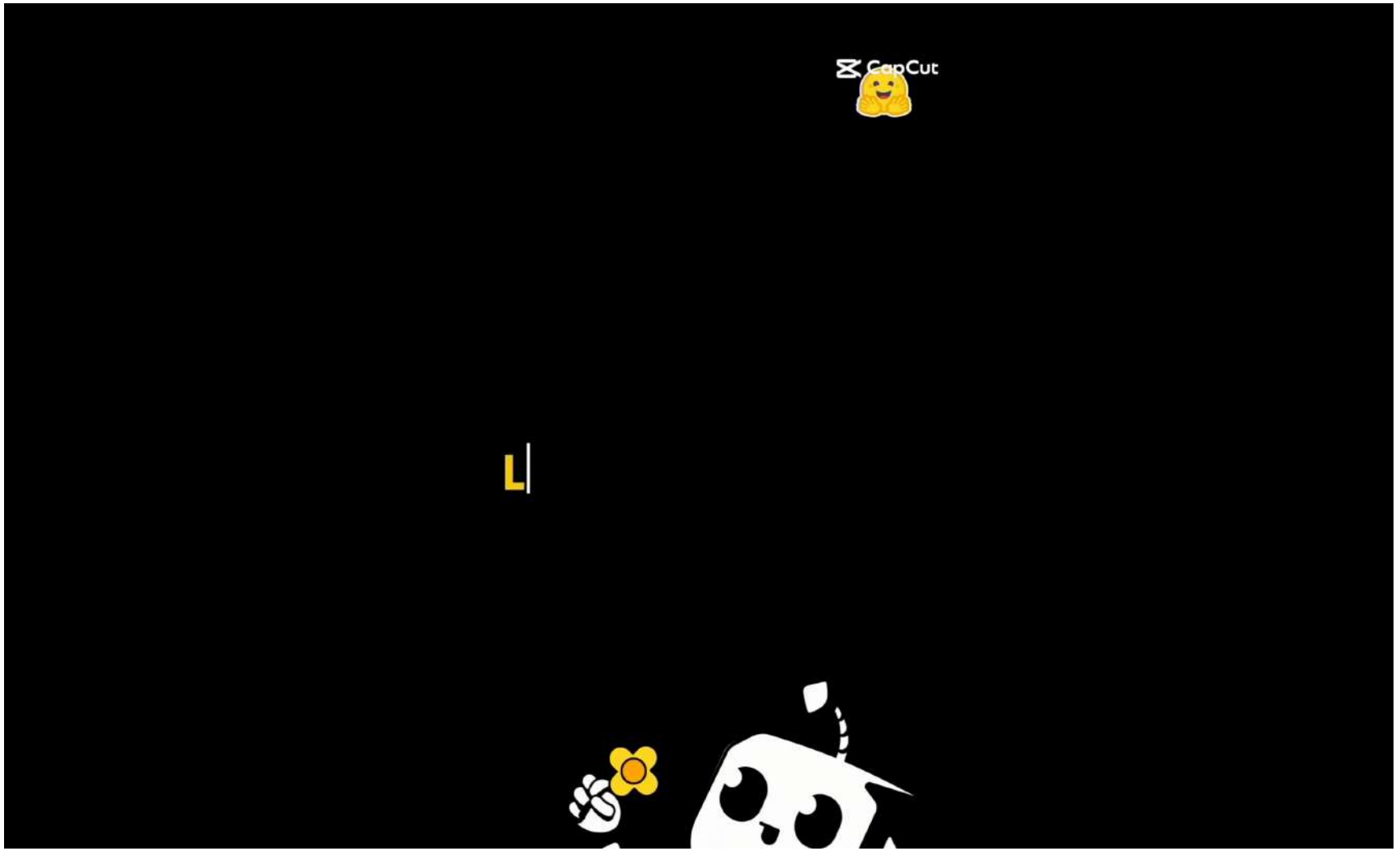
pretrained (fine tuning)

Deploy inferences (autonomous demonstrations)

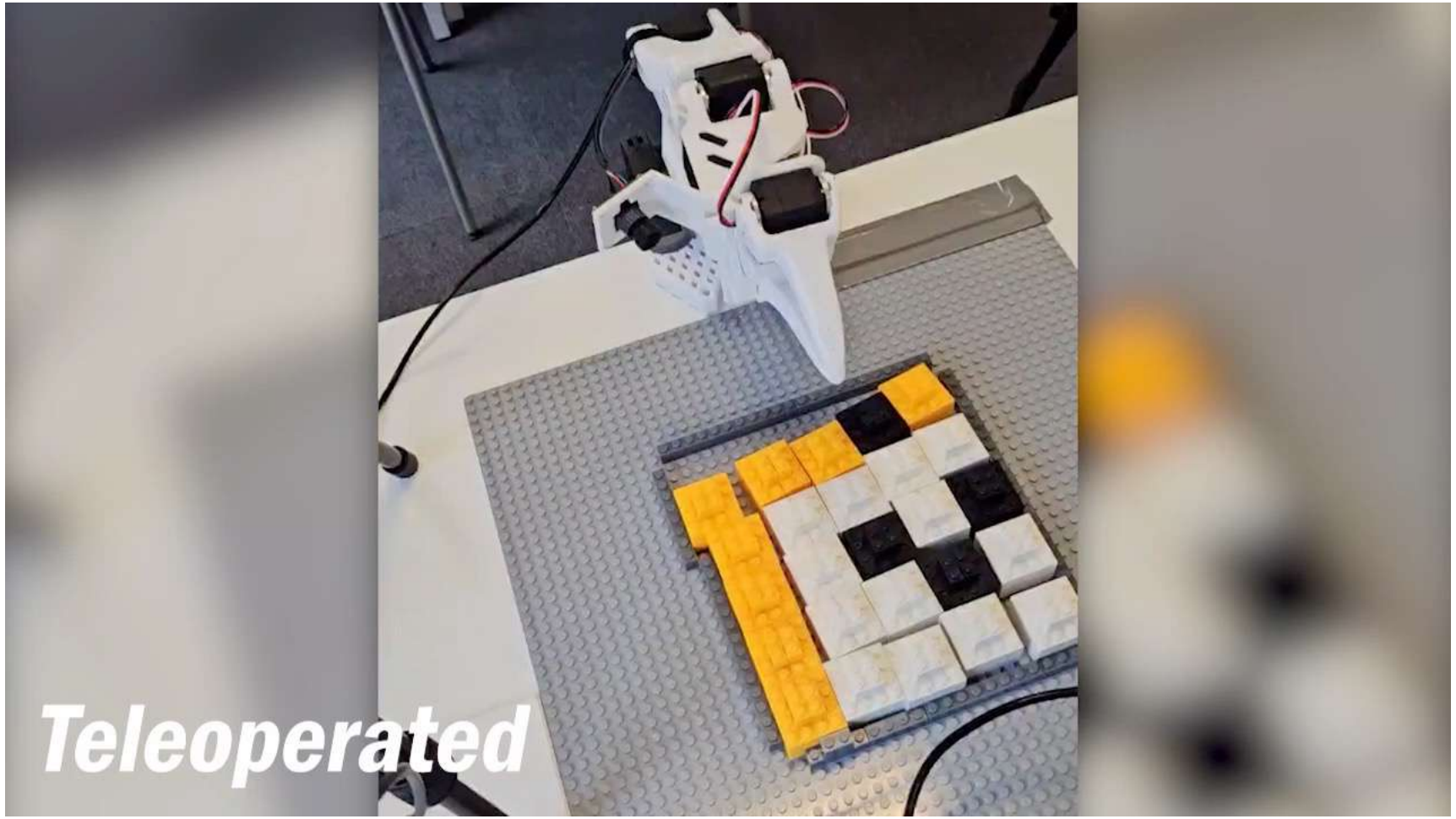


Source: https://huggingface.co/spaces/lerobot/robot-folding

The power of hackathons



The power of hackathons



The missing piece in robotics...

For humans, some tasks are incredibly hard without the sense of touch

Tactile sensors are important for precision tasks.

Today in a lot of robots, this tactile information is inferred from the vision data.





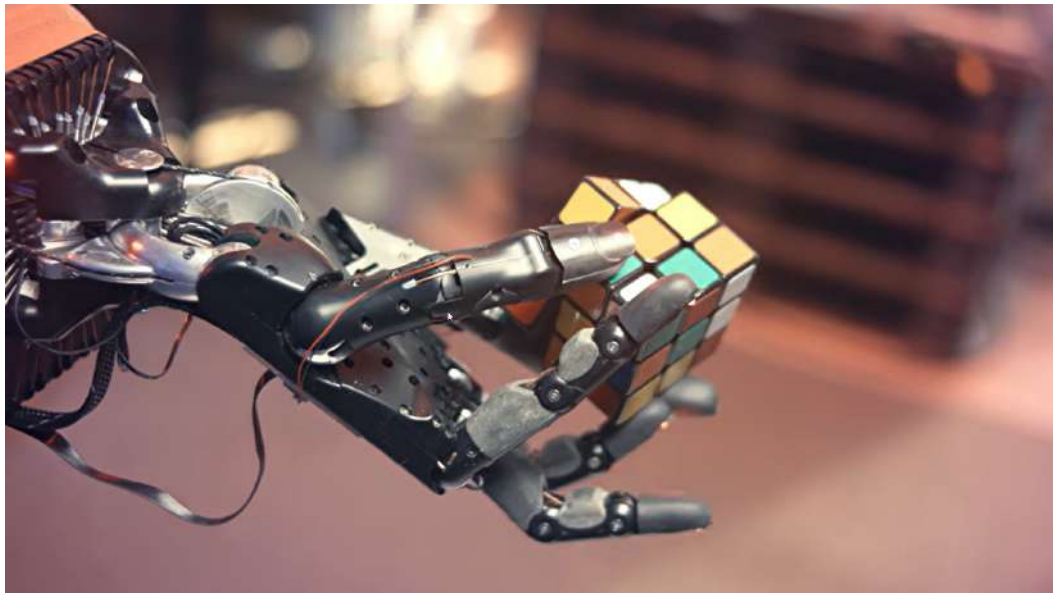


Back in 2018, OpenAI solves complex manipulation task

https://openai.com/index/learning-dexterity/

This was obtained without touch sensors
but with a lot of simulations





What surprised us

- **Tactile sensing is not necessary to manipulate real-world objects**. Our robot receives only the locations of the five fingertips along with the position and orientation of the cube. Although the robot hand has touch sensors on its fingertips, we didn't need to

video: https://www.youtube.com/watch?v=x4O8pojMF0w

Objectives

**How to give the sense of touch to a robot,
with a single point of sensing?**



Joseph Rigal



Caroline Pascal

Agnostic to the sensor technology

Wormsensing piezoelectric strain sensors,
but also, accelerometers, load cells, ... (7 technologies tested)

Agnostic to the acquisition system

From ~50 euros to 6000 euros high-fidelity acquisition systems

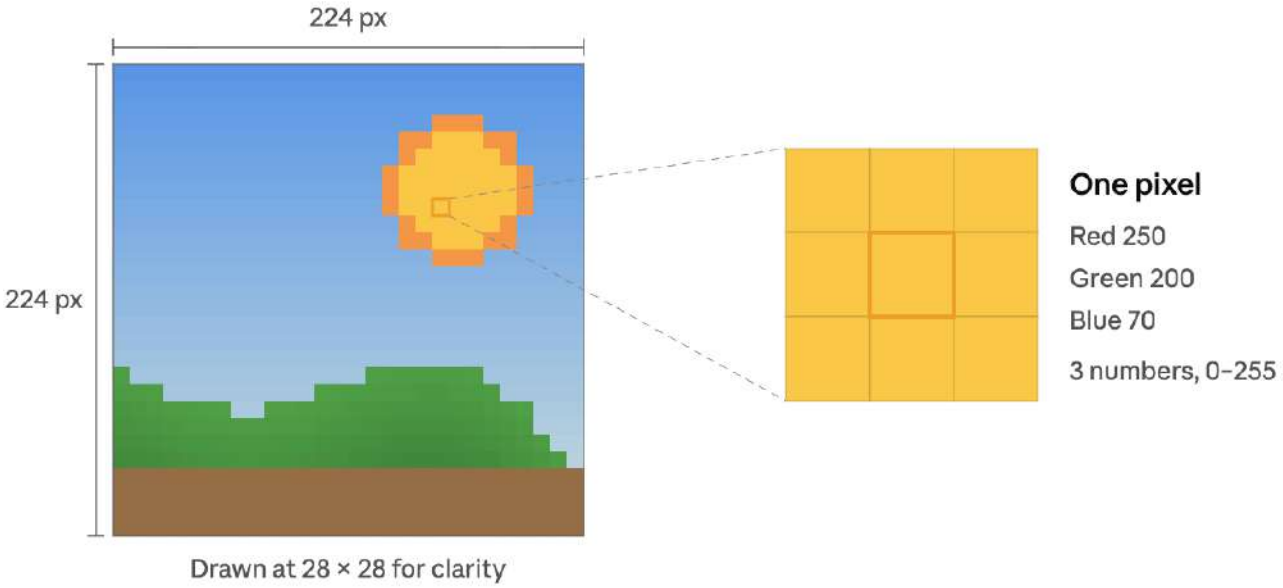
Agnostic to the model

Providing that it has a vision encoder

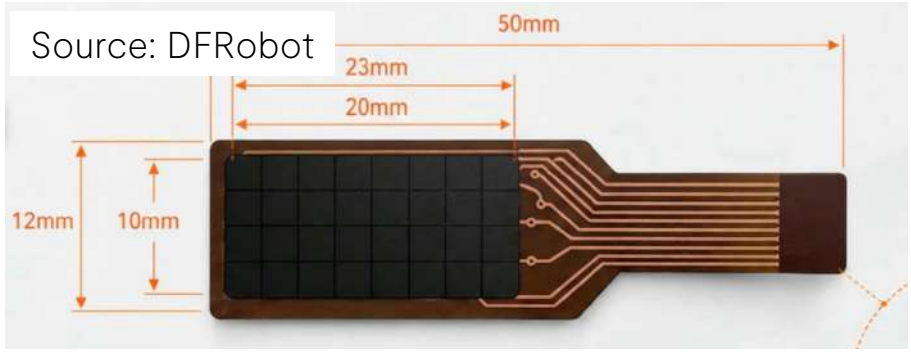

2020
Creation
24+
passionate people
10 M€
Total investment
80+
customers
60% international
worms
cea leti
Dragonfly®
product line-up
Qualified Sep. 2024
Hyperion
production line (Jan. 2024)
2 M units/y
Production capacity

SpectRobot

Typical webcam frame (30 fps)



What is typically used so far in robots



0	0	93	156	116	0	0	0
0	0	436	645	439	103	0	0
0	193	1130	1452	1116	421	0	0
0	132	874	1081	880	325	0	0

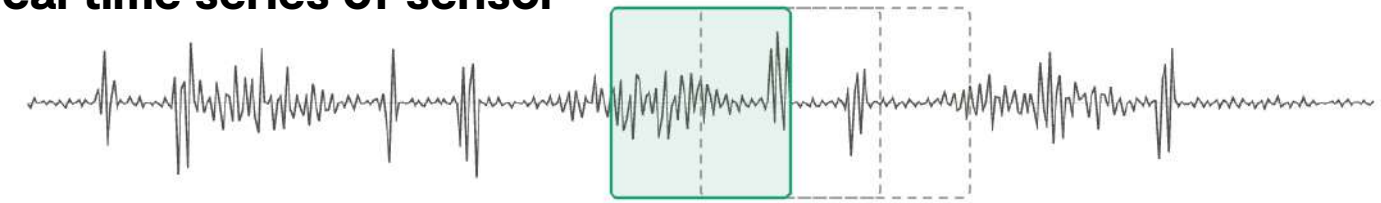
Very much like vision, purely spatial … and missing the dynamic contents (limited to 30 fps)

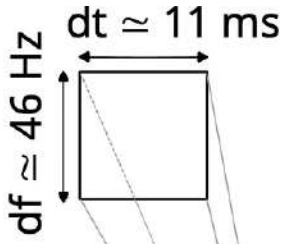
SpectRobot



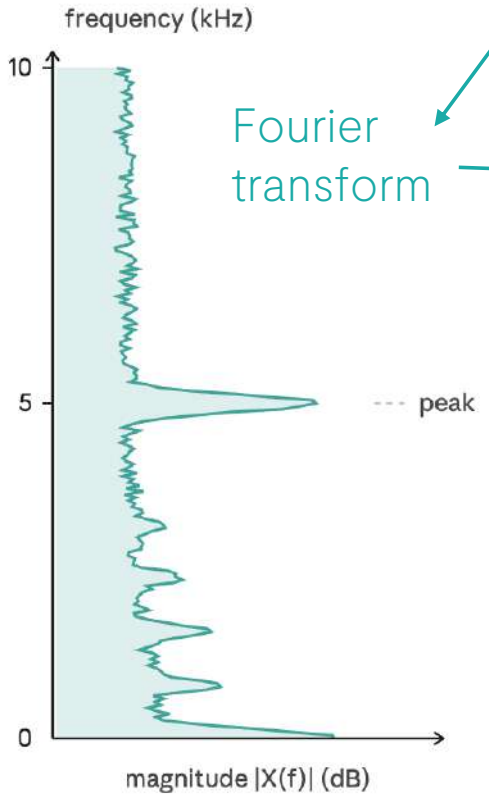
Industrial single-point sensors: strain sensors, accelerometers, load cells, ...

Typical time series of sensor



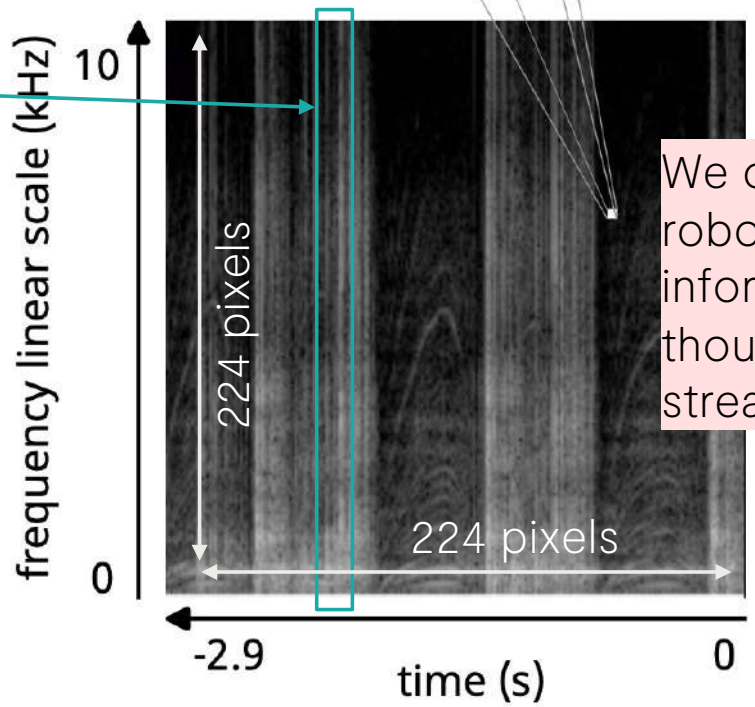


Spectrum



Fourier transform

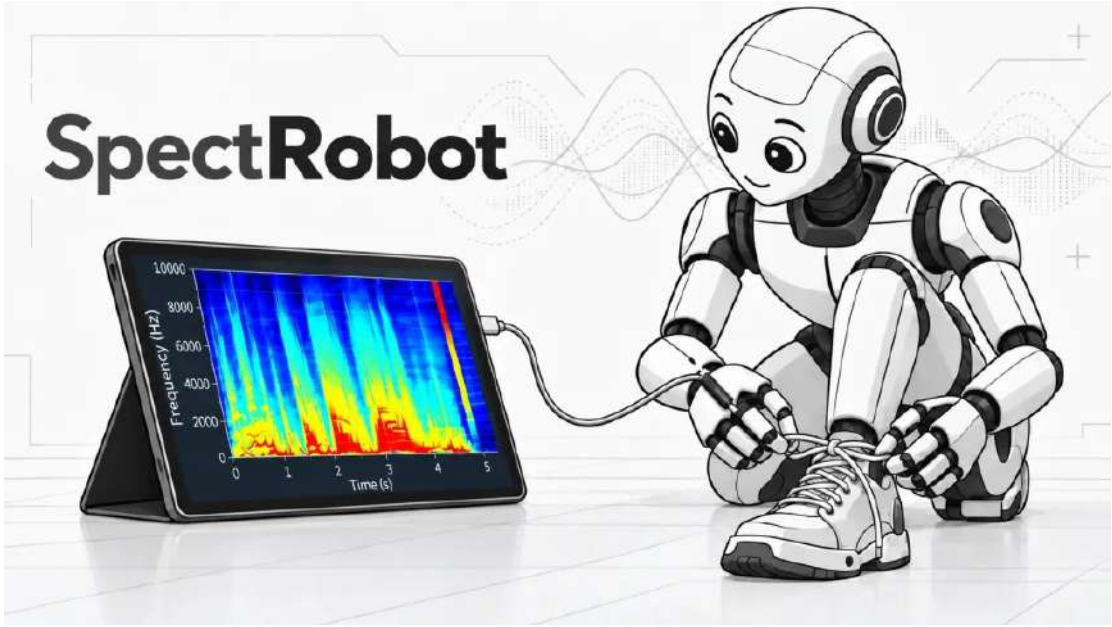
Spectrogram (short-time Fourier transform)



We can give to the robot a 100,000 Hz information, even though it is streamed at 30 fps!

SpectRobot

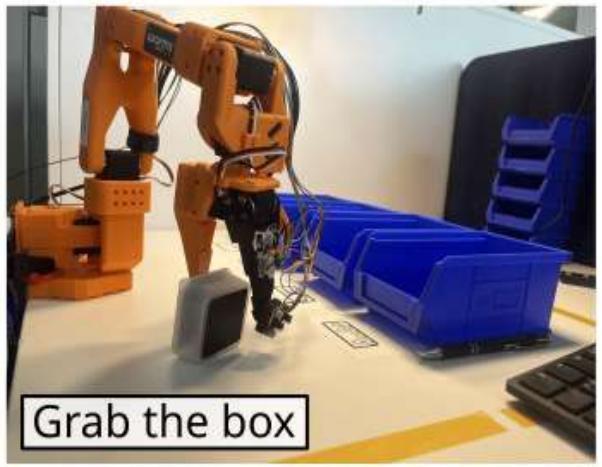
The project is fully open sourced at:
https://github.com/spectrobot-project/spectrobot



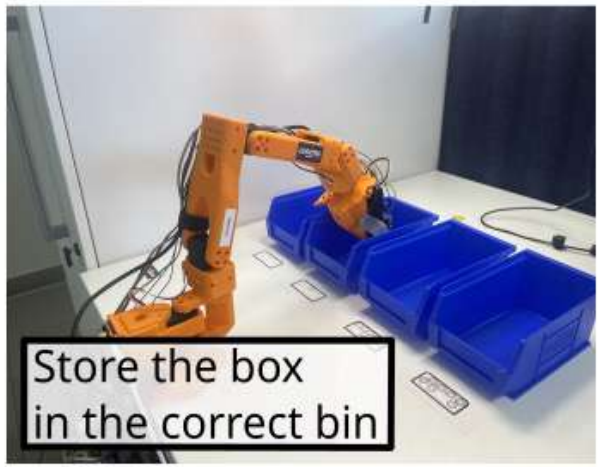


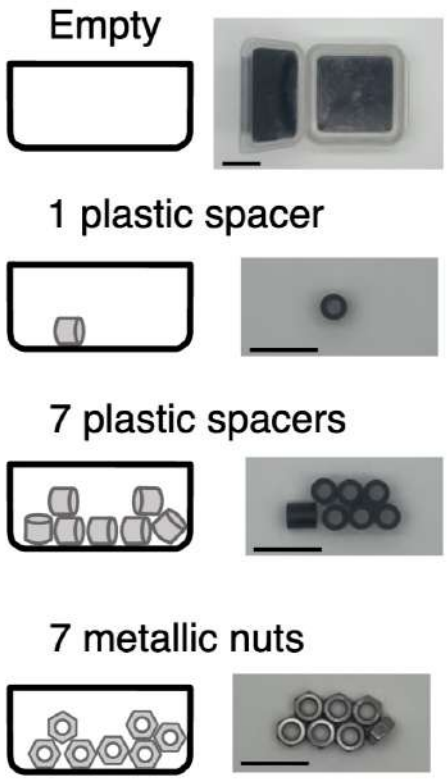
SpectRobot

Task: pick, shake and place an opaque box in the right bin, by guessing its content.



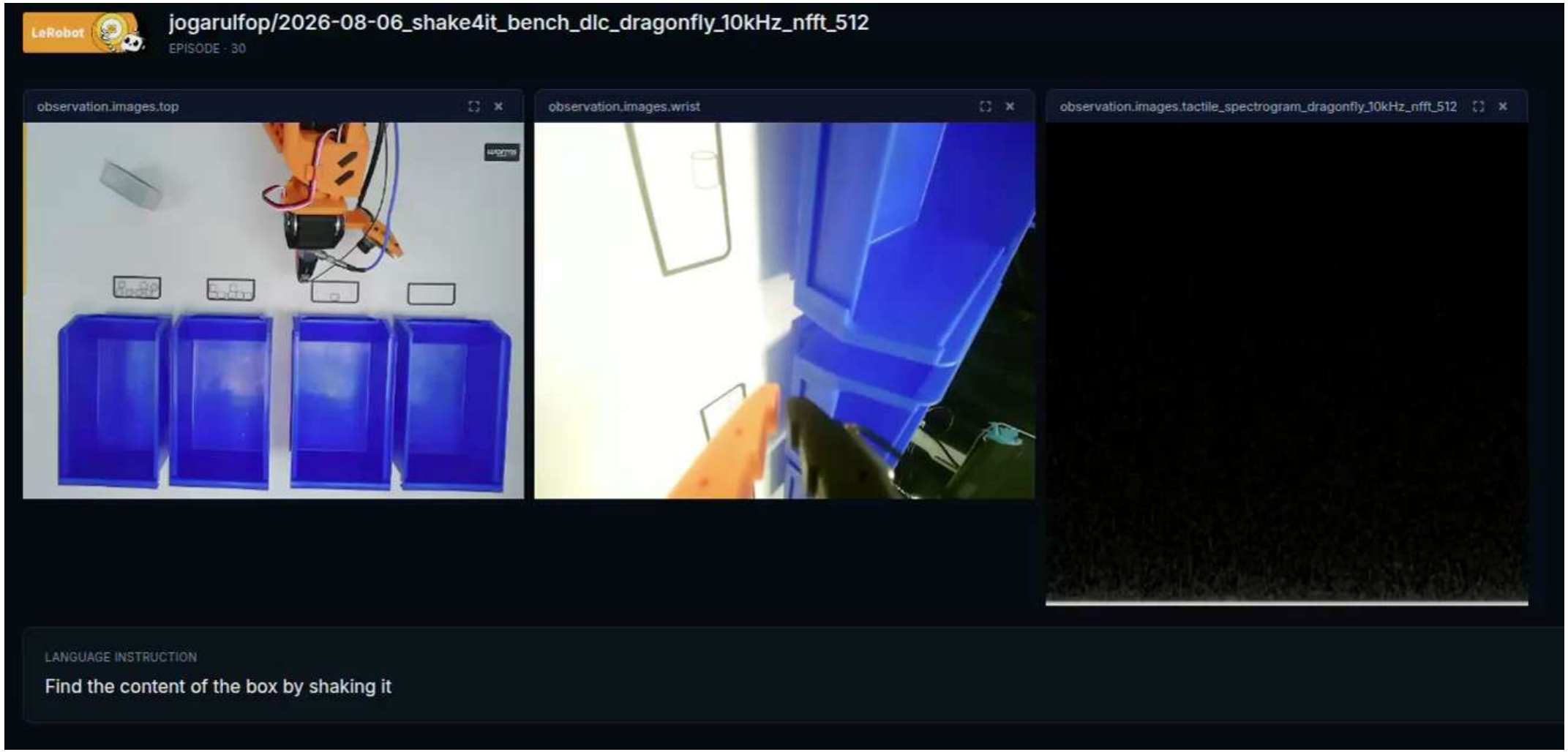






- Training with ACT model (Action Chunking with Transformers)
- Dataset of 40 episodes
- 100,000 steps lasting ~3h on a gaming computer

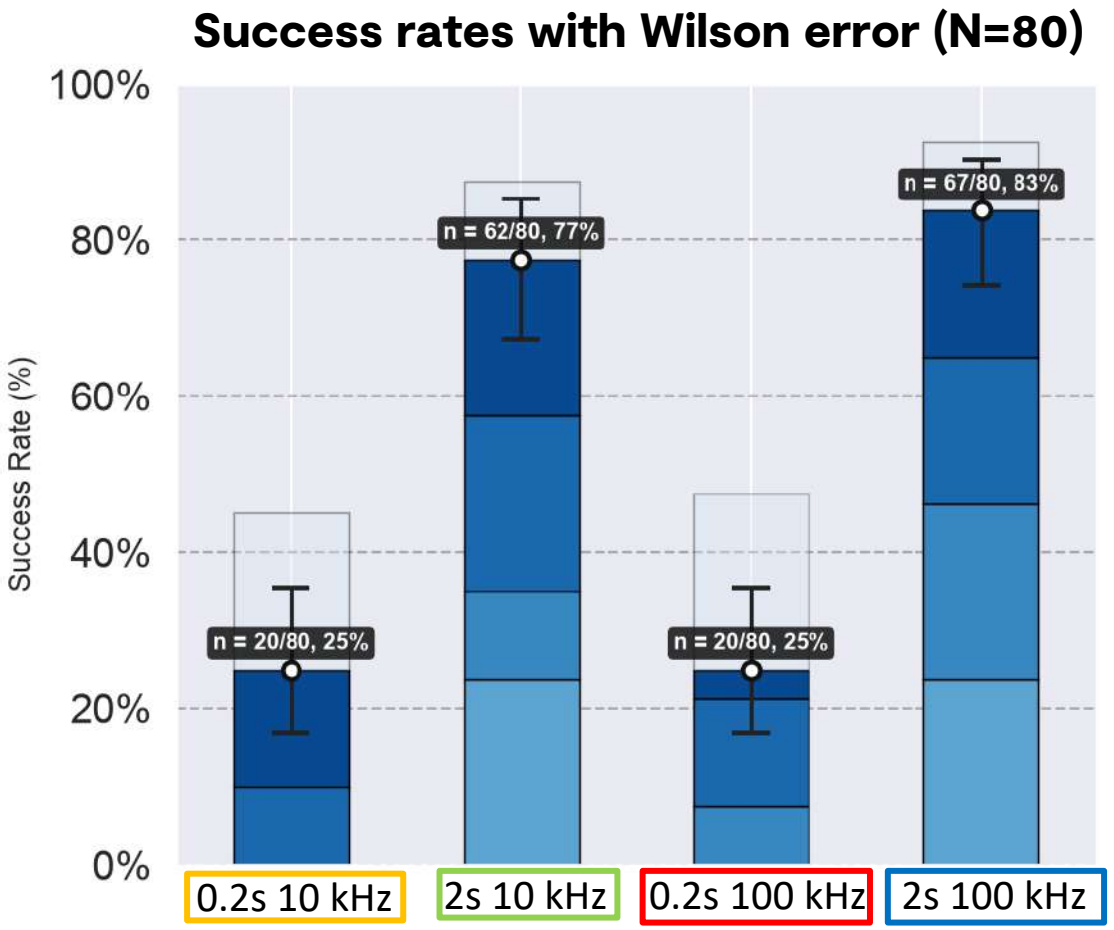
SpectRobot

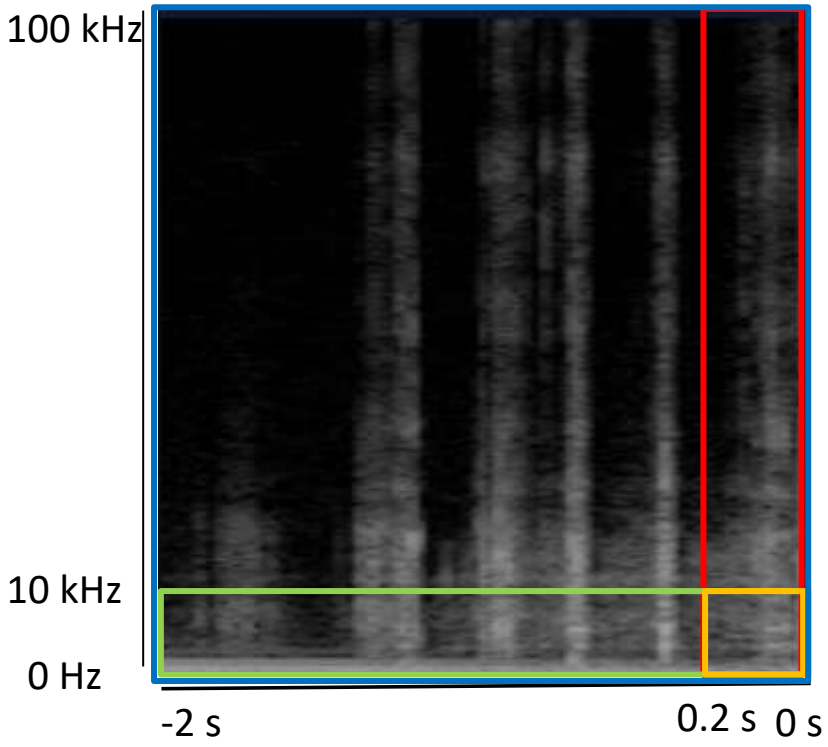

LeRobot
jogarulfop/2026-08-06_shake4it_bench_dlc_dragonfly_10kHz_nfft_512
EPISODE · 30
observation.images.top
observation.images.wrist
observation.images.tactile_spectrogram_dragonfly_10kHz_nfft_512
LANGUAGE INSTRUCTION
Find the content of the box by shaking it

Inference



SpectRobot

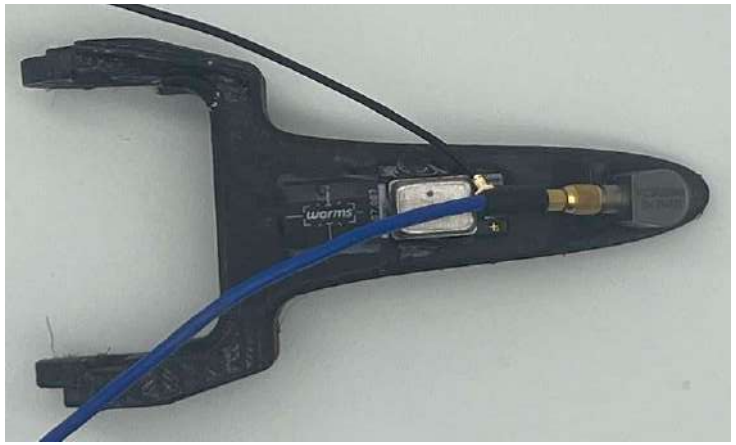




The history is very important for the success of the task

SpectRobot

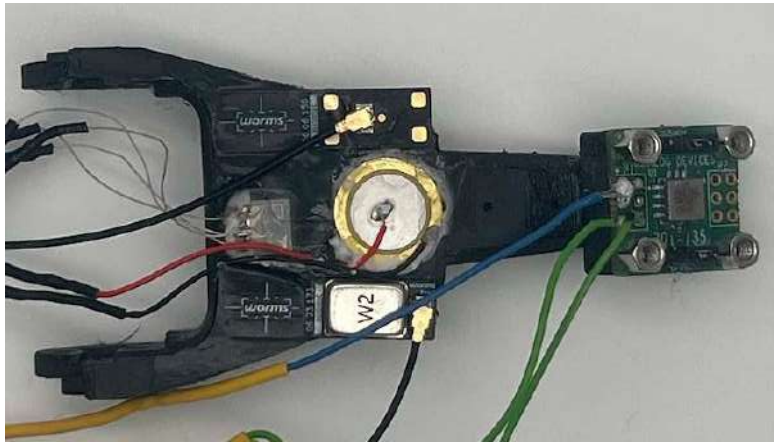
Tests with an accelerometer



Tests with a load cell



Tests with low-cost sensors





arXiv

Computer Science > Robotics

[Submitted on 21 Sep 2026 (v1), last revised 12 Oct 2026 (this version, v3)]

Learning tactile perception from high-bandwidth single-point sensing

Joseph Rigal, Emmanuel Virot, Caroline Pascal

Tactile sensing is increasingly being incorporated into learning-based robotic manipulation, yet many existing approaches rely on spatially distributed sensors. Here we introduce {SpectRobot}, a framework that transforms single-point tactile signals into compact time-frequency spectrograms. These spectrograms encode high-bandwidth tactile

https://arxiv.org/abs/2609.24621

Thank you!

emmanuel.virot@wormsensing.com

LeRobot tutorials
huggingface.co/docs/lerobot

SpectRobot webpage & GitHub
spectrobot-project.github.io

Paper "'Learning tactile perception from high-bandwidth single-point sensing"
arxiv.org/pdf/2609.24621

Thank you!

emmanuel.virot@wormsensing.com

Interested in organizing a Hackathon in Grenoble? (2027)



